

30) A says "I am the Knave," B says "I am the Knave," and C says "I am the Knave"

Soln: From the soln. of Problem 29, we know that only a spy can say that "I am the Knave". Since ~~exactly one~~ exactly one person can be a spy from the given constraints, so No solution exists in this situation.

31) A says "I am the Knight," B says "A is telling the truth," and C says "I am the spy."

Soln: Suppose, A is the Knight. It is consistent with the fact that a Knight always speaks the truth, since "I am the Knight" is True.
B says "A is telling the truth", which is true. $\therefore$ B must be the spy.
 $\therefore$ C is the Knave. C says "I am the spy" which is False and consistent with the fact that a Knave always lies.

Soln 1: $A \rightarrow \text{Knight}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knave}$

Suppose, A is the Knave. $\therefore$ "I am the Knight" is False, which is consistent with the fact that a Knave always lies.

B says "A is telling the truth" which is False. $\therefore$ B must be the spy.
 $\therefore$ C is the Knight. A Knight can never lie. C says ~~he~~ "I am the spy" which is False. This is a contradiction. $\therefore$ A cannot be the Knave

Suppose, A is the spy. $\therefore$ "I am the Knight" is False which is consistent with the fact that a spy sometimes lies.

B says "A is telling the truth" which is False. $\therefore$ B must be the Knave.
 $\therefore$ C is the Knight. But, a Knight can never lie. C says "I am the spy" which is False. This is a contradiction. $\therefore$ A cannot be the spy

Unique soln: $A \rightarrow \text{Knight}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knave}$